

# **WEEK – 1**

## **SOFTWARE REQUIREMENT SPECIFICATION**

### **Project Title**

**ShopNest – E-Commerce Order  
Management Database System**

**Submitted By,**

**S.Yuvasri**

**II-BCA-“B”**

| <b>S.No</b> | <b>Title</b>                |
|-------------|-----------------------------|
| 1           | Introduction                |
| 2           | Purpose                     |
| 3           | Scope                       |
| 4           | Overall System Description  |
| 5           | Functional Requirements     |
| 6           | Non-Functional Requirements |
| 7           | Assumptions                 |
| 8           | Constraints                 |
| 9           | Conclusion                  |

# 1. Introduction

The **ShopNest – E-Commerce Order Management Database System** is a centralized database solution developed to support the daily operations of an e-commerce platform. The system manages customer information, product records, inventory, suppliers, orders, payments, shipments, and customer reviews using a structured database. It ensures accurate data storage, minimizes redundancy, and supports efficient business operations.

## 2. Purpose

The purpose of this Software Requirement Specification (SRS) is to define the software requirements for the ShopNest – E-Commerce Order Management Database System. This document serves as the foundation for database design, Entity-Relationship (ER) modeling, normalization, SQL implementation, testing, and future maintenance.

## 3. Scope

The software will provide the following functionalities:

- Customer registration and information management.
- Product and category management.
- Supplier information management.
- Inventory monitoring and stock updates.
- Customer order processing.
- Order details management.
- Payment recording and payment status tracking.
- Shipment tracking and delivery status management.
- Customer review management.
- Business report generation.
- Centralized storage and retrieval of business data.

## 4. Overall System Description

The system is designed to maintain all e-commerce information in a centralized relational database using the following predefined entities:

- Customer
- Category
- Product
- Supplier

- Order
- Order Details
- Payment
- Shipment
- Review

The software allows authorized users to create, update, retrieve, and manage business data while maintaining consistency and integrity throughout the database.

## 5. Functional Requirements

The system shall provide the following functions:

### 5.1 Customer Management

- Register new customers.
- Store and maintain customer information.
- Update customer details when required.

### 5.2 Category Management

- Create and maintain product categories.
- Assign products to appropriate categories.

### 5.3 Product Management

- Add new products.
- Update product information.
- Maintain product prices and stock quantities.

### 5.4 Supplier Management

- Store supplier information.
- Maintain supplier contact details.

### 5.5 Inventory Management

- Monitor available stock.
- Update stock quantities after order processing.

### 5.6 Order Management

- Create customer orders.
- Store order information.
- Maintain order status.
- Record ordered products with quantity and unit price.

### 5.7 Payment Management

- Record payment details.

- Maintain payment status.
- Store payment method and payment date.

### 5.8 Shipment Management

- Record shipment information.
- Maintain delivery address.
- Update shipment and delivery status.

### 5.9 Review Management

- Store customer ratings.
- Record customer comments for products.

### 5.10 Report Generation

- Generate reports related to customers.
- Generate product reports.
- Generate order reports.
- Generate payment reports.
- Generate shipment reports.

## 6. Non-Functional Requirements

The system should satisfy the following quality requirements:

### Security

- Only authorized users shall access the database.
- Customer and business data shall be protected from unauthorized access.

### Reliability

- The system shall provide consistent and accurate data processing.

### Performance

- The system shall retrieve and update records with minimal response time.

### Availability

- The database shall be available whenever business operations are performed.

### Data Integrity

- The system shall maintain accurate and consistent data across all entities.

### Scalability

- The system shall support future business growth and increased data volume.

### Maintainability

- The software shall allow future modifications with minimal effort.

### Backup and Recovery

- Database backup and recovery mechanisms shall protect business data.

### Usability

- The system shall provide a simple and user-friendly interface.

### Compatibility

- The database shall operate with standard database management systems.

## 7. Assumptions

- Customers provide accurate registration details.
- Products are categorized correctly before being made available.
- Inventory records are updated after every order.
- Each order belongs to one registered customer.
- Payment information is entered correctly.
- Shipment information is updated after dispatch.
- Customer reviews are submitted only by registered customers.
- Authorized users follow proper system procedures.

## 8. Constraints

- Only the predefined core entities and attributes specified in the project are used.
- Entity names and attributes cannot be modified.
- The project focuses only on the database system.
- Online payment gateway integration is outside the project scope.
- Internet connectivity is required to access the system.
- Data accuracy depends on valid user input.

## 9. Conclusion

The Software Requirement Specification defines the functional and non-functional requirements of the **ShopNest – E-Commerce Order Management Database System**. It establishes a clear foundation for database design, implementation, testing, and future enhancements. By meeting these requirements, the system will support efficient e-commerce operations through secure, reliable, and centralized data management.